21: Model Comparison for Logistic Regression

Stat 230 - Spring 2026

```
library(dplyr)
library(broom)
library(ggformula)
```

In this activity we'll compare models for the fake news data using likelihood ratio tests (LRTs) and information criteria. To load the data, run the following code chunk:

```
news <- read.csv("https://aluby.github.io/stat230/data/fake_news_bayesrules.csv")
```

We'll start from the same model we used last time:

```
news_mod <- glm(type_num ~ title_words + title_has_excl + title_caps_percent,
                data = news, family = "binomial")

tidy(news_mod)
```

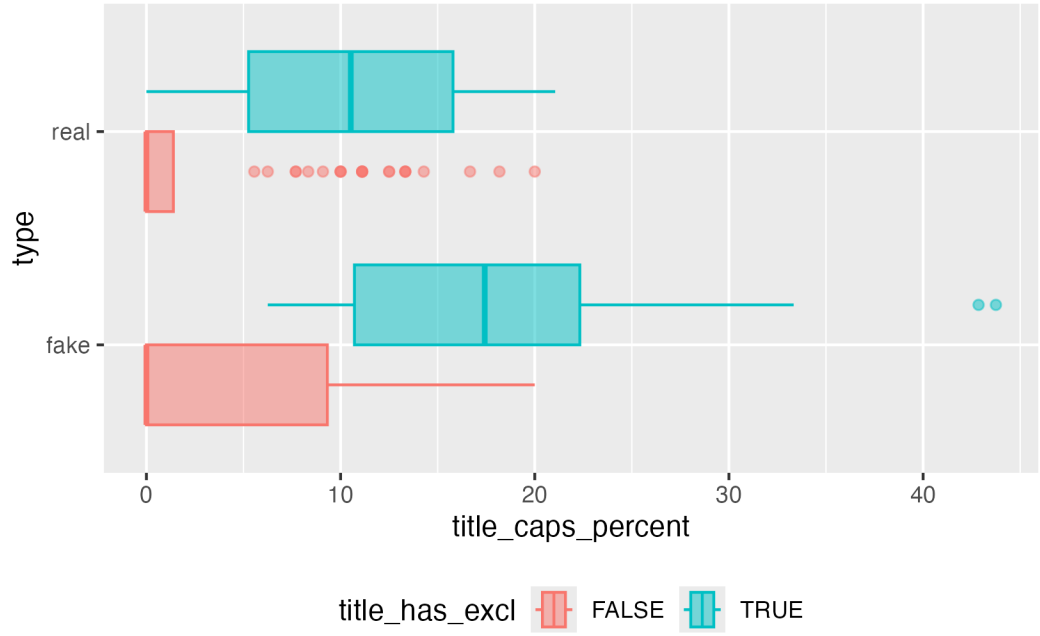
```
# A tibble: 4 x 5
  term               estimate std.error statistic  p.value
  <chr>                 <dbl>     <dbl>     <dbl>    <dbl>
1 (Intercept)          -2.23      0.672      -3.32 0.000894
2 title_words           0.120     0.0581      2.07 0.0383
3 title_has_exclTRUE    2.00      0.849       2.35 0.0186
4 title_caps_percent    0.0498    0.0295      1.69 0.0908
```

Our question today: **do we need an interaction between** `title_has_excl` **and** `title_caps_percent`**?** In other words, is the association between the percent of capital letters and the odds of being fake news different for titles with and without an exclamation point?

1 Part 1: Exploratory data analysis

First, a boxplot of `title_caps_percent` for real and fake articles, separately for titles with and without an exclamation point:

```
gf_boxplot(type ~ title_caps_percent, color = ~title_has_excl, fill = ~title_has_excl
           data = news, alpha = .5) +
  theme(legend.position = "bottom")
```



To check whether the slopes might differ, we also make an empirical logit plot with a line for each group. We bin `title_caps_percent` and calculate the empirical logit **within each bin and each group**.

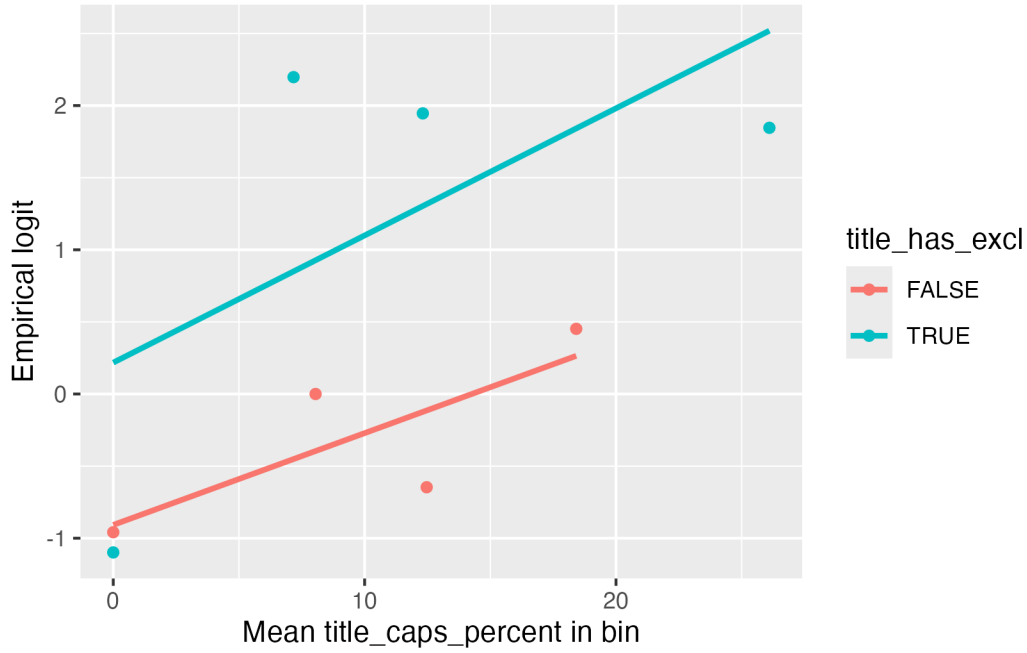
```
news_agg <- news |>
  mutate(tcp_bin = cut(title_caps_percent, c(-Inf, 0, 5, 10, 15, Inf))) |>
  group_by(tcp_bin, title_has_excl) |>
  summarize(mean_x = mean(title_caps_percent),
            n = n(),
            n_fake = sum(type_num),
            emp_logit = log((n_fake + 0.5) / (n - n_fake + 0.5)),
            .groups = "drop")

news_agg
```

```
# A tibble: 8 x 6
  tcp_bin   title_has_excl mean_x     n n_fake emp_logit
  <fct>     <lgl>           <dbl> <int>  <int>     <dbl>
1 (-Inf,0]  FALSE            0       91     25    -0.959
2 (-Inf,0]  TRUE             0        1      0    -1.10
3 (5,10]    FALSE            8.05    18      9     0
4 (5,10]    TRUE             7.17     4      4     2.20
5 (10,15]   FALSE           12.5     15      5    -0.647
6 (10,15]   TRUE            12.3      3      3     1.95
7 (15, Inf] FALSE           18.4      8      5     0.452
8 (15, Inf] TRUE            26.1     10      9     1.85
```

> **i** Why cut() and not ntile()?
>
> About 61% of the titles have exactly 0% capital letters, so equal-size bins would put most bins at the same x value. We create the bins by hand instead. (No titles fall in the (0, 5] bin, so it doesn't appear in the summary.)

```
gf_point(emp_logit ~ mean_x, color = ~title_has_excl, data = news_agg,
         xlab = "Mean title_caps_percent in bin", ylab = "Empirical logit") |>
  gf_smooth(method = "lm")
```



Q1. How many articles are in each of the cells of `news_agg`? Which group has very few observations?

Q2. Based on these plots, do you think we need an interaction (different slopes for the two groups)? How confident are you, given how little data there is for titles with exclamation points?

2 Part 2: The interaction model

Fill in the blanks to fit the model with an interaction between `title_has_excl` and `title_caps_percent` (and keep `title_words` as a main effect).

```
news_mod2 <- glm(type_num ~ title_words + ___ * ___,
                 data = news, family = "binomial")

tidy(news_mod2)
```

Q3. Write out the fitted model on the log-odds scale for titles **with** an exclamation point and for titles **without** one.

Q4. Interpret the coefficient for the interaction term `title_has_exclTRUE:title_caps_percent` on the **odds** scale. (*Hint:* $e^{\widehat{\beta}}$ is a ratio of odds ratios.)

3 Part 3: LRT for the interaction

The likelihood ratio test compares a **reduced** model to a **full** model that contains all of the reduced model's terms (plus some more). In R, we use `anova()` with `test = "LRT"`, and give the reduced model first.

```
anova(___, ___, test = "LRT")
```

Q5. Fill in the blanks to test whether we need the interaction term. Write out the null and alternative hypotheses, then report the test statistic (Λ, the `Deviance` column), the degrees of freedom, the p-value, and your conclusion in context.

Q6. The coefficient table from `tidy(news_mod2)` also contains a z-test for the interaction term. Compare its p-value to the LRT p-value. Are they close?

We can also verify that the LRT statistic is the **drop in deviance**. The deviance of a model is $-2 \log L$, and a smaller deviance means a better fit:

```r
deviance(news_mod)   # reduced model
deviance(news_mod2)  # full model
deviance(news_mod) - deviance(news_mod2)
```

4 Part 4: Do we need `title_has_excl` at all?

The variable `title_has_excl` appears in **two** terms of the interaction model (the main effect and the interaction), so to ask whether we need it at all we must remove both. Fit this reduced model and use an LRT to compare it to news_mod2:

```r
reduced_model <- glm(type_num ~ title_words + title_caps_percent,
                     data = news, family = "binomial")

anova(reduced_model, news_mod2, test = "LRT")
```

5 Part 5: The overall test

The output of `summary()` gives us the deviance of the model we fit (the residual deviance) and the deviance of an intercept-only model (the null deviance).

```r
summary(news_mod)
```

Q9. Use the null deviance and residual deviance to calculate the LRT statistic for $H_0 : \beta_1 = \beta_2 = \beta_3 = 0$ (all slopes are 0) by hand. Then use `pchisq()` to find the p-value. The degrees of freedom is the difference in the two models' degrees of freedom.

```r
lambda <- ___
df <- ___
1 - pchisq(lambda, df)
```

Check your work by fitting an intercept-only model and comparing:

```r
null_model <- glm(type_num ~ 1, data = news, family = "binomial")
anova(null_model, news_mod, test = "LRT")
```

6 Part 6: AIC and BIC

```
AIC(reduced_model, news_mod, news_mod2)
BIC(reduced_model, news_mod, news_mod2)
```

Q10. Which of the three models do AIC and BIC prefer? Do they agree with the LRTs? Which model would you use to explain the fake news data?

7 Your turn

The dataset also has the sentiment of the article: `negative` and `positive` (the percent of words with negative and positive sentiment).

- Fit a model that adds both variables to `news_mod`.
- Use an LRT to test whether the two sentiment variables help ($d = 2$).
- Compare the LRT result to the individual z-tests for `negative` and `positive`.
- Compare AIC and BIC for the two models. Do the criteria agree? Which model would you choose, and why?

```
# your code here
```

---

8 Function quick reference

Function	Purpose
`glm(y ~ x, data, family = "binomial")`	Fit a binary logistic regression model
`cut(x, breaks)`	Create bins from specific cut points
`anova(reduced, full, test = "LRT")`	Likelihood ratio test comparing two nested models
`deviance(model)`	Deviance ($-2 \log L$) of a fitted model
`pchisq(x, df)`	Probability from a χ^2 distribution (use `1 - pchisq()` for a p-value)

Function	Purpose
`AIC()`, `BIC()`	Information criteria for one or more fitted models (smaller is better)